

Red Hat OpenShift Data Foundation 4.22

Deploying OpenShift Data Foundation using IBM Power

Instructions on deploying Red Hat OpenShift Data Foundation on IBM Power

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Abstract

Read this document for instructions about how to install Red Hat OpenShift Data Foundation to use local storage on IBM Power.

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PREFACE

This document describes how to deploy Red Hat OpenShift Data Foundation using IBM Power.

PROVIDING FEEDBACK ON RED HAT DOCUMENTATION

We appreciate your input on our documentation. Do let us know how we can make it better.

To give feedback, create a Jira ticket:

Procedure

1. Log in to the Jira.
2. Click **Create** in the top navigation bar
3. Enter a descriptive title in the Summary field.
4. Enter your suggestion for improvement in the Description field. Include links to the relevant parts of the documentation.
5. Select **Documentation** in the Components field.
6. Click Create at the bottom of the dialogue.

PREFACE

Red Hat OpenShift Data Foundation supports deployment on existing Red Hat OpenShift Container Platform (RHOCP) IBM Power clusters in connected or disconnected environments along with out-of-the-box support for proxy environments.

Both internal and external OpenShift Data Foundation clusters are supported on IBM Power. See the Planning your deployment guide and the Preparing to deploy OpenShift Data Foundation chapter for more information about deployment requirements.

To deploy OpenShift Data Foundation, follow the appropriate deployment process based on your requirement:

- Internal-Attached Devices mode – Deploy using local storage devices, or Deploy standalone Multicloud Object Gateway component
- External mode using Red Hat Ceph Storage

Additional resources

- Planning your deployment
- Preparing to deploy OpenShift Data Foundation
- Deploy standalone Multicloud Object Gateway component
- External mode

CHAPTER 1. PREPARING TO DEPLOY OPENSHIFT DATA FOUNDATION

When you deploy OpenShift Data Foundation on OpenShift Container Platform using the local storage devices provided by IBM Power, you can create internal cluster resources. This approach internally provisions base services so that all the applications can access additional storage classes..

Before you begin the deployment of Red Hat OpenShift Data Foundation using a local storage, ensure that you meet the resource requirements. See Requirements for installing OpenShift Data Foundation using local storage devices.

- Optional: If you want to enable cluster-wide encryption using the external Key Management System (KMS) follow these steps:
    - Ensure that you have a valid Red Hat OpenShift Data Foundation Advanced subscription. To know how subscriptions for OpenShift Data Foundation work, see knowledgebase article on OpenShift Data Foundation subscriptions.
    - When the Token authentication method is selected for encryption then refer to Enabling cluster-wide encryption with the Token authentication using KMS.
    - When the Kubernetes authentication method is selected for encryption then refer to Enabling cluster-wide encryption with the Kubernetes authentication using KMS .
    - Ensure that you are using signed certificates on your Vault servers.

After you have addressed the above, perform the following steps:

1. Install Local Storage Operator .
2. Install the Red Hat OpenShift Data Foundation Operator .
3. Find available storage devices .
4. Create OpenShift Data Foundation cluster on IBM Power .

1.1. REQUIREMENTS FOR INSTALLING OPENSHIFT DATA FOUNDATION USING LOCAL STORAGE DEVICES

Before installing OpenShift Data Foundation on IBM Power Systems using local storage devices, your cluster must meet specific node, disk, and resource requirements.

Node requirements

- The cluster must consist of at least three OpenShift Container Platform worker nodes in the cluster with locally attached storage devices on each of them.
    - Each of the three selected nodes must have at least one raw block device available to be used by OpenShift Data Foundation.
    - The devices to be used must be empty, that is, there should be no persistent volumes (PVs), volume groups (VGs), or local volumes (LVs) remaining on the disks.
- You must have a minimum of three labeled worker nodes.

- Each node that has local storage devices to be used by OpenShift Data Foundation must have a specific label to deploy OpenShift Data Foundation pods. To label the nodes, use the following command:

```
$ oc label nodes <NodeNames> cluster.ocs.openshift.io/openshift-storage=''
```

For more information, see the *Resource requirements* section in the Planning guide.

Disconnected environment requirements

For the list of packages to be mirrored for installing OpenShift Data Foundation in disconnected environments, see the Disconnected environment section in the *Planning your deployment* guide.

Disaster recovery requirements

Disaster Recovery features supported by Red Hat OpenShift Data Foundation require all of the following prerequisites to successfully implement a disaster recovery solution:

- A valid Red Hat OpenShift Data Foundation Advanced subscription.
- A valid Red Hat Advanced Cluster Management (RHACM) for Kubernetes subscription.

To know in detail how subscriptions for OpenShift Data Foundation work, see the knowledgebase article on OpenShift Data Foundation subscriptions.

For detailed disaster recovery solution requirements, see the *Configuring OpenShift Data Foundation Disaster Recovery for OpenShift Workloads* guide, and the *Requirements and recommendations* section of the Install guide in Red Hat Advanced Cluster Management for Kubernetes documentation.

Additional resources

- Planning guide - Disconnected environment
- Planning guide - Resource requirements
- Knowledgebase article on OpenShift Data Foundation subscriptions
- Configuring OpenShift Data Foundation Disaster Recovery for OpenShift Workloads
- Red Hat Advanced Cluster Management for Kubernetes - Requirements and recommendations

CHAPTER 2. DEPLOY OPENSHIFT DATA FOUNDATION USING LOCAL STORAGE DEVICES

Use this section to deploy OpenShift Data Foundation on IBM Power infrastructure where OpenShift Container Platform is already installed.

Also, it is possible to deploy only the Multicloud Object Gateway (MCG) component with OpenShift Data Foundation. For more information, see Deploy standalone Multicloud Object Gateway .

Perform the following steps to deploy OpenShift Data Foundation:

1. Install the Local Storage Operator .
2. Install the Red Hat OpenShift Data Foundation Operator .
3. Find available storage devices .
4. Create an OpenShift Data Foundation cluster on IBM Power .

2.1. INSTALLING LOCAL STORAGE OPERATOR

Use this procedure to install the Local Storage Operator from the Software Catalog before creating OpenShift Data Foundation clusters on local storage devices.

Procedure

1. Log in to the OpenShift Web Console.
2. Click **Ecosystem → Software Catalog**
3. Type **local storage** in the **Filter by keyword...** box to find the **Local Storage Operator** from the list of operators and click on it.
4. Set the following options on the **Install Operator** page:
   a. Update channel as **stable**.
   b. Installation Mode as **A specific namespace on the cluster**
   c. Installed Namespace as **Operator recommended namespace openshift-local-storage**.
   d. Approval Strategy as **Automatic**.
5. Click **Install**.

Verification

- Verify that the Local Storage Operator shows a green tick indicating successful installation.

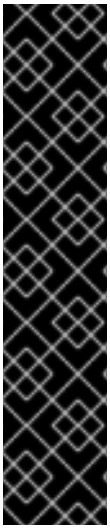
2.2. INSTALLING RED HAT OPENSHIFT DATA FOUNDATION OPERATOR

You can install Red Hat OpenShift Data Foundation Operator using the Red Hat OpenShift Container Platform Software Catalog.

For information about the hardware and software requirements, see Planning your deployment.

Prerequisites

- Access to an OpenShift Container Platform cluster using an account with **cluster-admin** and Operator installation permissions.
- You must have at least three worker nodes in the Red Hat OpenShift Container Platform cluster.



IMPORTANT

- When you need to override the cluster-wide default node selector for OpenShift Data Foundation, you can use the following command in the command line interface to specify a blank node selector for the **openshift-storage** namespace (create openshift-storage namespace in this case):

```
$ oc annotate namespace openshift-storage openshift.io/node-selector=
```

- Taint a node as **infra** to ensure only Red Hat OpenShift Data Foundation resources are scheduled on that node. This helps you save on subscription costs. For more information, see How to use dedicated worker nodes for Red Hat OpenShift Data Foundation chapter in the Managing and Allocating Storage Resources guide.

Procedure

1. Log in to the OpenShift Web Console.
2. Click **Ecosystem → Software Catalog**
3. Scroll or type **OpenShift Data Foundation** into the **Filter by keyword** box to find the **OpenShift Data Foundation** Operator.
4. Click **Install**.
5. Set the following options on the **Install Operator** page:
   a. Update Channel as **stable-4.22**.
   b. Installation Mode as **A specific namespace on the cluster**.
   c. Installed Namespace as **Operator recommended namespace openshift-storage**. If Namespace **openshift-storage** does not exist, it is created during the operator installation.
6. Select **Approval Strategy** as **Automatic** or **Manual**.
   If you select **Automatic** updates, then the Operator Lifecycle Manager (OLM) automatically upgrades the running instance of your Operator without any intervention.

   If you select **Manual** updates, then the OLM creates an update request. As a cluster administrator, you must then manually approve that update request to update the Operator to a newer version.
7. Ensure that the **Enable** option is selected for the **Console plugin**.

8. Click **Install**.

Verification

- Verify that the **OpenShift Data Foundation** Operator shows a green tick indicating successful installation.
- After the operator is successfully installed, the web console automatically reloads to apply the changes. During this process, a temporary error message might appear on the page and this is expected and disappears after the refresh completes.
    - In the Web Console, navigate to **Storage** and verify if **Data Foundation** is available.

2.3. ENABLING CLUSTER-WIDE ENCRYPTION WITH KMS USING THE TOKEN AUTHENTICATION METHOD

You can enable the key value backend path and policy in the vault for token authentication.

Prerequisites

- Administrator access to the vault.
- A valid Red Hat OpenShift Data Foundation Advanced subscription. For more information, see the knowledgebase article on OpenShift Data Foundation subscriptions.
- Carefully, select a unique path name as the backend **path** that follows the naming convention since you cannot change it later.

Procedure

1. Enable the Key/Value (KV) backend path in the vault.
   For vault KV secret engine API, version 1:

   ```
   $ vault secrets enable -path=odf kv
   ```

   For vault KV secret engine API, version 2:

   ```
   $ vault secrets enable -path=odf kv-v2
   ```

2. Create a policy to restrict the users to perform a write or delete operation on the secret:

   ```
   echo '
   path "odf/*" {
     capabilities = ["create", "read", "update", "delete", "list"]
   }
   path "sys/mounts" {
   capabilities = ["read"]
   }'| vault policy write odf -
   ```

3. Create a token that matches the above policy:

   ```
   $ vault token create -policy=odf -format json
   ```

2.4. ENABLING CLUSTER-WIDE ENCRYPTION WITH KMS USING THE KUBERNETES AUTHENTICATION METHOD

You can enable the Kubernetes authentication method for cluster-wide encryption using the Key Management System (KMS).

Prerequisites

- Administrator access to Vault.
- A valid Red Hat OpenShift Data Foundation Advanced subscription. For more information, see the knowledgebase article on OpenShift Data Foundation subscriptions.
- The OpenShift Data Foundation operator must be installed from the Software Catalog.
- Select a unique path name as the backend **path** that follows the naming convention carefully. You cannot change this path name later.

Procedure

1. Create a service account:

```
$ oc -n openshift-storage create serviceaccount <serviceaccount_name>
```

where, ***<serviceaccount_name>*** specifies the name of the service account.

For example:

```
$ oc -n openshift-storage create serviceaccount odf-vault-auth
```

2. Create **clusterrolebindings** and **clusterroles**:

```
$ oc -n openshift-storage create clusterrolebinding vault-tokenreview-binding --
clusterrole=system:auth-delegator --serviceaccount=openshift-
storage:_<serviceaccount_name>_
```

For example:

```
$ oc -n openshift-storage create clusterrolebinding vault-tokenreview-binding --
clusterrole=system:auth-delegator --serviceaccount=openshift-storage:odf-vault-auth
```

3. Create a secret for the **serviceaccount** token and CA certificate.

```
$ cat <<EOF | oc create -f -
apiVersion: v1
kind: Secret
metadata:
  name: odf-vault-auth-token
  namespace: openshift-storage
  annotations:
    kubernetes.io/service-account.name: <serviceaccount_name>
```

```
type: kubernetes.io/service-account-token
data: {}
EOF
```

where, ***<serviceaccount_name>*** is the service account created in the earlier step.

4. Get the token and the CA certificate from the secret.

```
$ SA_JWT_TOKEN=$(oc -n openshift-storage get secret odf-vault-auth-token -o jsonpath="
{.data['token']}" | base64 --decode; echo)
$ SA_CA_CRT=$(oc -n openshift-storage get secret odf-vault-auth-token -o jsonpath="
{.data['ca\.crt']}" | base64 --decode; echo)
```

5. Retrieve the OCP cluster endpoint.

```
$ OCP_HOST=$(oc config view --minify --flatten -o jsonpath="{.clusters[0].cluster.server}")
```

6. Fetch the service account issuer:

```
$ oc proxy &
$ proxy_pid=$!
$ issuer="$( curl --silent http://127.0.0.1:8001/.well-known/openid-configuration | jq -r
.issuer)"
$ AUDIENCE=$(echo '{"apiVersion": "authentication.k8s.io/v1", "kind": "TokenRequest"}' |
kubectl create f --raw /api/v1/namespaces/default/serviceaccounts/default/token | jq -r
'.spec.audiences' | jq -r .[0])
$ kill $proxy_pid
```

Or, for Vault version 1.21.0 and higher:

```
$ oc proxy &
$ proxy_pid=$!
$ ISSUER=$(echo '{"apiVersion": "authentication.k8s.io/v1", "kind": "TokenRequest"}' \   |
kubectl create f --raw /api/v1/namespaces/default/serviceaccounts/default/token \   | jq -r
'.status.token' \   | cut -d . -f2 \   | base64 -d \   | jq '.iss')
$ AUDIENCE=$(echo '{"apiVersion": "authentication.k8s.io/v1", "kind": "TokenRequest"}' |
kubectl create f --raw /api/v1/namespaces/default/serviceaccounts/default/token | jq -r
'.spec.audiences' | jq -r .[0])
$ kill $proxy_pid
```

7. Use the information collected in the previous step to setup the Kubernetes authentication method in Vault:

```
$ vault auth enable kubernetes
```

```
$ vault write auth/kubernetes/config \
        token_reviewer_jwt="$SA_JWT_TOKEN" \
        kubernetes_host="$OCP_HOST" \
        kubernetes_ca_cert="$SA_CA_CRT" \
        issuer="$issuer"
```

Or, for Vault version 1.21.0 and higher:

```
$ oc exec -n vault -ti vault-0 – \
   vault write auth/kubernetes/config \
   token_reviewer_jwt="$SA_JWT_TOKEN" \
   kubernetes_host="$OCP_HOST" \
   kubernetes_ca_cert="$SA_CA_CRT" \
   issuer="$ISSUER"\
   token_audience="$AUDIENCE"
```



IMPORTANT

To configure the Kubernetes authentication method in Vault when the issuer is empty:

```
$ vault write auth/kubernetes/config \
        token_reviewer_jwt="$SA_JWT_TOKEN" \
        kubernetes_host="$OCP_HOST" \
        kubernetes_ca_cert="$SA_CA_CRT"
```

8. Enable the Key/Value (KV) backend path in Vault.
   For Vault KV secret engine API, version 1:

```
$ vault secrets enable -path=odf kv
```

   For Vault KV secret engine API, version 2:

```
$ vault secrets enable -path=odf kv-v2
```

9. Create a policy to restrict the users to perform a **write** or **delete** operation on the secret:

```
echo '
path "odf/*" {
  capabilities = ["create", "read", "update", "delete", "list"]
}
path "sys/mounts" {
capabilities = ["read"]
}'| vault policy write odf -
```

10. Generate the roles:

```
$ vault write auth/kubernetes/role/odf-rook-ceph-op \
      bound_service_account_names=rook-ceph-system,rook-ceph-osd,noobaa \
      bound_service_account_namespaces=openshift-storage \
      policies=odf \
      ttl=1440h
```

    The role **odf-rook-ceph-op** is later used while you configure the KMS connection details during the creation of the storage system.

```
$ vault write auth/kubernetes/role/odf-rook-ceph-osd \
      bound_service_account_names=rook-ceph-osd \
      bound_service_account_namespaces=openshift-storage \
```

```
policies=odf \
ttl=1440h
```

2.4.1. Enabling and disabling key rotation when using KMS

Security common practices require periodic encryption of key rotation. You can enable or disable key rotation when using KMS.

2.4.1.1. Enforcing precedence for key rotation

Configure key rotation precedence to control whether the rotation schedule is read from the PVC annotation or the storage class in OpenShift Data Foundation.

In key rotation, precedence refers to the order in which the system checks for scheduled annotations. In OpenShift Data Foundation, the default precedence is set to storage class (recommended). This means the system reads annotations only from the storage class.

However, if you want the system to check the persistent volume claim (PVC) first and then fall back to the storage class, you can configure this behavior by setting the **schedule-precedence** to PVC in the CSI-addons ConfigMap.

Procedure

1. Define the **ConfigMap** for csi-addons to set PVC precedence:

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: csi-addons-config
  namespace: openshift-storage
data:
  "schedule-precedence": "pvc"
```

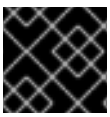
2. Restart the **csi-addons** operator pod for the changes to take effect:

```
oc delete po -n openshift-storage -l "app.kubernetes.io/name=csi-addons"
```

2.4.1.2. Enabling key rotation

Enable key rotation to automatically re-encrypt persistent volume data at a scheduled interval using the CSI Addons framework in OpenShift Data Foundation.

To enable key rotation, add the annotation **keyrotation.csiaddons.openshift.io/schedule: <value>** to a **PersistentVolumeClaim**, **Namespace**, or **StorageClass** (in decreasing order of precedence).



IMPORTANT

Key rotation is only supported for RBD backed volumes.

2.4.1.2.1. Annotating a namespace for key rotation

Add the key rotation annotation to a namespace so that all RBD-backed persistent volumes in that namespace are re-encrypted on the configured schedule.

Prerequisites

- Key rotation is only supported for RBD backed volumes.

Procedure

1. Verify the namespace exists:

```
$ oc get namespace default
NAME      STATUS   AGE
default   Active   5d2h
```

2. Annotate the namespace with the desired rotation schedule:

```
$ oc annotate namespace default "keyrotation.csiaddons.openshift.io/schedule=@weekly"
namespace/default annotated
```

<value> can be **@hourly**, **@daily**, **@weekly**, **@monthly**, or **@yearly**. If **<value>** is empty, the default is **@weekly**.



NOTE

- When any schedule for key rotation is used, including **@daily**, **@weekly**, **@monthly**, and cron expressions such as **0 * * * ***, multiple CronJobs are created. To prevent all Jobs from running simultaneously and overloading high-scale systems, executions are staggered within a 2-hour window from the scheduled start time (e.g., **@weekly** would be 00:00, 00:05, 00:10, and so on).
    - If the configured schedule interval is less than 2 hours, the stagger window is reduced to match the interval. For example, a job running every 30 minutes will be staggered within a 00:00–00:30 window rather than 00:00–02:00.

2.4.1.2.2. Annotating a storage class for key rotation

Add the key rotation annotation to a storage class so that all RBD-backed persistent volumes provisioned from that storage class are re-encrypted on the configured schedule.

Prerequisites

- Key rotation is only supported for RBD backed volumes.

Procedure

1. Verify the storage class exists:

```
$ oc get storageclass rbd-sc
NAME      PROVISIONER        RECLAIMPOLICY   VOLUMEBINDINGMODE
ALLOWVOLUMEEXPANSION   AGE
rbd-sc    rbd.csi.ceph.com   Delete          Immediate           true                   5d2h
```

2. Annotate the storage class with the desired rotation schedule:

```
$ oc annotate storageclass rbd-sc "keyrotation.csiaddons.openshift.io/schedule=@weekly"
storageclass.storage.k8s.io/rbd-sc annotated
```

<value> can be **@hourly**, **@daily**, **@weekly**, **@monthly**, or **@yearly**. If **<value>** is empty, the default is **@weekly**.

2.4.1.2.3. Annotating a PersistentVolumeClaim for key rotation

Add the key rotation annotation directly to a PersistentVolumeClaim to re-encrypt that volume on the configured schedule, overriding any namespace or storage class annotation.

Prerequisites

- Key rotation is only supported for RBD backed volumes.

Procedure

1. Verify the PersistentVolumeClaim exists:

```
$ oc get pvc data-pvc
NAME      STATUS   VOLUME                                     CAPACITY   ACCESS MODES
STORAGECLASS      AGE
data-pvc  Bound    pvc-f37b8582-4b04-4676-88dd-e1b95c6abf74   1Gi        RWO
default           20h
```

2. Annotate the PVC with the desired rotation schedule:

```
$ oc annotate pvc data-pvc "keyrotation.csiaddons.openshift.io/schedule=@weekly"
persistentvolumeclaim/data-pvc annotated
```

<value> can be **@hourly**, **@daily**, **@weekly**, **@monthly**, or **@yearly**. If **<value>** is empty, the default is **@weekly**.

3. Verify the encryption key rotation cron job was created:

```
$ oc get encryptionkeyrotationcronjobs.csiaddons.openshift.io
NAME                 SCHEDULE    SUSPEND   ACTIVE   LASTSCHEDULE   AGE
data-pvc-1642663516  @weekly                                       3s
```

4. Optional: Update the schedule by overwriting the annotation:

```
$ oc annotate pvc data-pvc "keyrotation.csiaddons.openshift.io/schedule=*/1 * * * *" --
overwrite=true
persistentvolumeclaim/data-pvc annotated
```

5. Verify the updated cron job schedule:

```
$ oc get encryptionkeyrotationcronjobs.csiaddons.openshift.io
NAME                SCHEDULE    SUSPEND   ACTIVE   LASTSCHEDULE   AGE
data-pvc-1642664617  */1 * * * *                                   3s
```

2.4.1.3. Disabling key rotation

Disable automatic key rotation for all PVCs in a storage class or for a specific persistent volume claim in OpenShift Data Foundation.

You can disable key rotation for the following:

- All the persistent volume claims (PVCs) of storage class
- A specific PVC

Disabling key rotation for all PVCs of a storage class

To disable key rotation for all PVCs, update the annotation of the storage class:

```
$ oc get storageclass rbd-sc
NAME      PROVISIONER        RECLAIMPOLICY   VOLUMEBINDINGMODE
ALLOWVOLUMEEXPANSION   AGE
rbd-sc    rbd.csi.ceph.com   Delete          Immediate          true                5d2h
```

```
$ oc annotate storageclass rbd-sc "keyrotation.csiaddons.openshift.io/enable: false"
storageclass.storage.k8s.io/rbd-sc annotated
```

Disabling key rotation for a specific persistent volume claim

Procedure

1. Identify the **EncryptionKeyRotationCronJob** CR for the PVC you want to disable key rotation on:

   ```
   $ oc get encryptionkeyrotationcronjob -o jsonpath='{range .items[?
   (@.spec.jobTemplate.spec.target.persistentVolumeClaim=="<PVC_NAME>")]}
   {.metadata.name}{"\n"}{end}'
   ```

   Where **<PVC_NAME>** is the name of the PVC that you want to disable.

2. Apply the following to the **EncryptionKeyRotationCronJob** CR from the previous step to disable the key rotation:

   a. Update the **csiaddons.openshift.io/state** annotation from **managed** to **unmanaged**:

   ```
   $ oc annotate encryptionkeyrotationcronjob <encryptionkeyrotationcronjob_name>
   "csiaddons.openshift.io/state=unmanaged" --overwrite=true
   ```

   Where **<encryptionkeyrotationcronjob_name>** is the name of the **EncryptionKeyRotationCronJob** CR.

   b. Add **suspend: true** under the **spec** field:

   ```
   $ oc patch encryptionkeyrotationcronjob <encryptionkeyrotationcronjob_name> -p
   '{"spec": {"suspend": true}}' --type=merge.
   ```

   The key rotation will be disabled for the PVC.

2.5. FINDING AVAILABLE STORAGE DEVICES

Use this procedure to identify the device names for each of the three or more worker nodes that you have labeled with the OpenShift Data Foundation label **cluster.ocs.openshift.io/openshift-storage="** before creating PVs for IBM Power.

Procedure

1. List and verify the name of the worker nodes with the OpenShift Data Foundation label.

```
$ oc get nodes -l cluster.ocs.openshift.io/openshift-storage=
```

Example output:

```
NAME       STATUS   ROLES    AGE     VERSION
worker-0   Ready    worker   2d11h   v1.23.3+e419edf
worker-1   Ready    worker   2d11h   v1.23.3+e419edf
worker-2   Ready    worker   2d11h   v1.23.3+e419edf
```

2. Log in to each worker node that is used for OpenShift Data Foundation resources and find the name of the additional disk that you have attached while deploying Openshift Container Platform.

```
$ oc debug node/<node name>
```

Example output:

```
$ oc debug node/worker-0
Starting pod/worker-0-debug ...
To use host binaries, run `chroot /host`
Pod IP: 192.168.0.63
If you don't see a command prompt, try pressing enter.
sh-4.4#
sh-4.4# chroot /host
sh-4.4# lsblk
NAME   MAJ:MIN RM   SIZE RO TYPE MOUNTPOINT
loop1    7:1    0   500G  0 loop
sda      8:0    0   500G  0 disk
sdb      8:16   0   120G  0 disk
|-sdb1   8:17   0     4M  0 part
|-sdb3   8:19   0   384M  0 part
`-sdb4   8:20   0 119.6G  0 part
sdc      8:32   0   500G  0 disk
sdd      8:48   0   120G  0 disk
|-sdd1   8:49   0     4M  0 part
|-sdd3   8:51   0   384M  0 part
`-sdd4   8:52   0 119.6G  0 part
sde      8:64   0   500G  0 disk
sdf      8:80   0   120G  0 disk
|-sdf1   8:81   0     4M  0 part
|-sdf3   8:83   0   384M  0 part
`-sdf4   8:84   0 119.6G  0 part
sdg      8:96   0   500G  0 disk
sdh      8:112  0   120G  0 disk
|-sdh1   8:113  0     4M  0 part
|-sdh3   8:115  0   384M  0 part
`-sdh4   8:116  0 119.6G  0 part
```

```
sdi      8:128  0   500G  0 disk
sdj      8:144  0   120G  0 disk
|-sdj1   8:145  0     4M  0 part
|-sdj3   8:147  0   384M  0 part
`-sdj4   8:148  0 119.6G  0 part
sdk      8:160  0   500G  0 disk
sdl      8:176  0   120G  0 disk
|-sdl1   8:177  0     4M  0 part
|-sdl3   8:179  0   384M  0 part
`-sdl4   8:180  0 119.6G  0 part /sysroot
sdm      8:192  0   500G  0 disk
sdn      8:208  0   120G  0 disk
|-sdn1   8:209  0     4M  0 part
|-sdn3   8:211  0   384M  0 part /boot
`-sdn4   8:212  0 119.6G  0 part
sdo      8:224  0   500G  0 disk
sdp      8:240  0   120G  0 disk
|-sdp1   8:241  0     4M  0 part
|-sdp3   8:243  0   384M  0 part
`-sdp4   8:244  0 119.6G  0 part
```

In this example, for worker-0, the available local devices of 500G are **sda**, **sdc**, **sde**,**sdg**, **sdi**, **sdk**, **sdm**,**sdo**.

3. Repeat the above step for all the other worker nodes that have the storage devices to be used by OpenShift Data Foundation. See this Knowledge Base article for more details.

2.6. CREATING OPENSHIFT DATA FOUNDATION CLUSTER ON IBM POWER

Use this procedure to create an OpenShift Data Foundation cluster after you install the OpenShift Data Foundation operator.

Prerequisites

- Ensure that all the requirements in the Requirements for installing OpenShift Data Foundation using local storage devices section are met.
- You must have a minimum of three worker nodes with the same storage type and size attached to each node (for example, 200 GB SSD) to use local storage devices on IBM Power.
- Verify your OpenShift Container Platform worker nodes are labeled for OpenShift Data Foundation:

```
oc get nodes -l cluster.ocs.openshift.io/openshift-storage -o jsonpath='{range .items[*]}
{.metadata.name}{"\n"}'
```

To identify storage devices on each node, refer to Finding available storage devices.

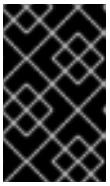
Procedure

1. Log into the OpenShift Web Console.

2. In **openshift-local-storage** namespace Click **Ecosystem → Installed Operators** to view the installed operators.
3. Click the **Local Storage** installed operator.
4. On the **Operator Details** page, click the **Local Volume** link.
5. Click **Create Local Volume**.
6. Click on **YAML view** for configuring Local Volume.
7. Define a **LocalVolume** custom resource for block PVs using the following YAML.

```
apiVersion: local.storage.openshift.io/v1
kind: LocalVolume
metadata:
  name: localblock
  namespace: openshift-local-storage
spec:
  logLevel: Normal
  managementState: Managed
  nodeSelector:
    nodeSelectorTerms:
      - matchExpressions:
          - key: kubernetes.io/hostname
            operator: In
            values:
              - worker-0
              - worker-1
              - worker-2
  storageClassDevices:
    - devicePaths:
        - /dev/sda
      storageClassName: localblock
      volumeMode: Block
```

The above definition selects **sda** local device from the **worker-0**, **worker-1** and **worker-2** nodes. The **localblock** storage class is created and persistent volumes are provisioned from **sda**.



IMPORTANT

Specify appropriate values of nodeSelector as per your environment. The device name should be same on all the worker nodes. You can also specify more than one devicePaths.

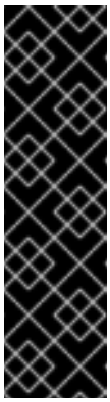
8. Click **Create**.
9. Confirm whether **diskmaker-manager** pods and **Persistent Volumes** are created.
   a. For **Pods**
      i. Click **Workloads → Pods** from the left pane of the OpenShift Web Console.
      ii. Select **openshift-local-storage** from the **Project** drop-down list.

iii. Check if there are **diskmaker-manager** pods for each of the worker node that you used while creating LocalVolume CR.

b. For **Persistent Volumes**

i. Click **Storage → PersistentVolumes** from the left pane of the OpenShift Web Console.

ii. Check the Persistent Volumes with the name **local-pv-***. Number of Persistent Volumes will be equivalent to the product of number of worker nodes and number of storage devices provisioned while creating localVolume CR.



IMPORTANT

- The flexible scaling feature is enabled only when the storage cluster that you created with three or more nodes are spread across fewer than the minimum requirement of three availability zones.
  For information about flexible scaling, see knowledgebase article on *Scaling OpenShift Data Foundation cluster using YAML when flexible scaling is enabled*.
- Flexible scaling features get enabled at the time of deployment and can not be enabled or disabled later on.

10. In the OpenShift Web Console, click **Storage → Data Foundation → Storage Systems → Create StorageSystem**.

11. In the Backing storage page, perform the following:

a. Select the **Use an existing StorageClass** option.

b. Select the required **Storage Class** that you used while installing LocalVolume.
By default, it is set to **none**.

c. Optional: Select **Use Ceph RBD as the default StorageClass** This avoids having to manually annotate a StorageClass.

d. Optional: Select **Use external PostgreSQL** checkbox to use an external PostgreSQL **[Technology preview]**.
This provides high availability solution for Multicloud Object Gateway where the PostgreSQL pod is a single point of failure.



IMPORTANT

OpenShift Data Foundation ships PostgreSQL images maintained by Red Hat, which are used to store metadata for the Multicloud Object Gateway. This PostgreSQL usage is at the application level.

As a result, OpenShift Data Foundation does not perform database-level optimizations or in-depth insights.

If customers have their own PostgreSQL that is well-maintained and optimized, we recommend using it. OpenShift Data Foundation supports external PostgreSQL instances.

Any PostgreSQL-related issues requiring code changes or deep technical analysis may need to be addressed upstream. This could result in longer resolution times.

i. Provide the following connection details:

- **Username**
- **Password**
- **Server name** and **Port**
- **Database name**

ii. Select **Enable TLS/SSL** checkbox to enable encryption for the Postgres server.

e. Click **Next**.

12. In the **Capacity and nodes** page, configure the following:

a. **Available raw capacity** is populated with the capacity value based on all the attached disks associated with the storage class. This takes some time to show up. The **Selected nodes** list shows the nodes based on the storage class.

b. In the **Configure performance** section, select one of the following performance profiles:

- Lean
  Use this in a resource constrained environment with minimum resources that are lower than the recommended. This profile minimizes resource consumption by allocating fewer CPUs and less memory.

- Balanced (default)
  Use this when recommended resources are available. This profile provides a balance between resource consumption and performance for diverse workloads.

- Performance
  Use this in an environment with sufficient resources to get the best performance. This profile is tailored for high performance by allocating ample memory and CPUs to ensure optimal execution of demanding workloads.



NOTE

You have the option to configure the performance profile even after the deployment using the **Configure performance** option from the options menu of the **StorageSystems** tab.



IMPORTANT

Before selecting a resource profile, make sure to check the current availability of resources within the cluster. Opting for a higher resource profile in a cluster with insufficient resources might lead to installation failures.

For more information about resource requirements, see Resource requirement for performance profiles.

c. Optional: Select the **Taint nodes** checkbox to dedicate the selected nodes for OpenShift Data Foundation.

d. Click **Next**.

13. Optional: In the **Security and network** page, configure the following based on your requirements:

a. To enable encryption, select **Enable data encryption for block and file storage**

i. Select either one or both the encryption levels:

- **Cluster-wide encryption**
  Encrypts the entire cluster (block and file).
- **StorageClass encryption**
  Creates encrypted persistent volume (block only) using encryption enabled storage class.

ii. Optional: Select the **Connect to an external key management service** checkbox. This is optional for cluster-wide encryption.

A. From the **Key Management Service Provider** drop-down list, either select **Vault** or **Thales CipherTrust Manager (using KMIP)**. If you selected **Vault**, go to the next step. If you selected **Thales CipherTrust Manager (using KMIP)**, go to step iii.

B. Select an **Authentication Method**.

Using Token authentication method

- Enter a unique **Connection Name**, host **Address** of the Vault server ('https://<hostname or ip>'), **Port** number and **Token**.
- Expand **Advanced Settings** to enter additional settings and certificate details based on your **Vault** configuration:
  - Enter the Key Value secret path in **Backend Path** that is dedicated and unique to OpenShift Data Foundation.
  - Optional: Enter **TLS Server Name** and **Vault Enterprise Namespace**.

- Upload the respective PEM encoded certificate file to provide the **CA Certificate**, **Client Certificate** and **Client Private Key** .
- Click **Save** and skip to step iv.

Using Kubernetes authentication method

- Enter a unique Vault **Connection Name**, host **Address** of the Vault server ('https://<hostname or ip>'), **Port** number and **Role** name.
- Expand **Advanced Settings** to enter additional settings and certificate details based on your **Vault** configuration:
  - Enter the Key Value secret path in **Backend Path** that is dedicated and unique to OpenShift Data Foundation.
  - Optional: Enter **TLS Server Name** and **Authentication Path** if applicable.
  - Upload the respective PEM encoded certificate file to provide the **CA Certificate**, **Client Certificate** and **Client Private Key** .
  - Click **Save** and skip to step iv.

C. To use **Thales CipherTrust Manager (using KMIP)** as the KMS provider, follow the steps below:

I. Enter a unique **Connection Name** for the Key Management service within the project.

II. In the **Address** and **Port** sections, enter the IP of Thales CipherTrust Manager and the port where the KMIP interface is enabled. For example:

- **Address**: 123.34.3.2
- **Port**: 5696

III. Upload the **Client Certificate**, **CA certificate**, and **Client Private Key**.

IV. If StorageClass encryption is enabled, enter the Unique Identifier to be used for encryption and decryption generated above.

V. The **TLS Server** field is optional and used when there is no DNS entry for the KMIP endpoint. For example, **kmip_all_<port>.ciphertrustmanager.local**.

D. Select a **Network**.

I. Select **Default (OVN)** network as Multus is not yet supported on OpenShift Data Foundation on IBM Power.

E. Click **Next**.

b. To enable in-transit encryption, select **In-transit encryption**.

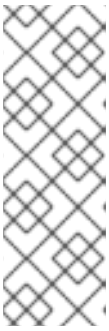
i. Select a **Network**.

ii. Click **Next**.

14. In the **Review and create** page::

    a. Review the configurations details. To modify any configuration settings, click **Back** to go back to the previous configuration page.

    b. Click **Create StorageSystem**.



NOTE

When your deployment has five or more nodes, racks, or rooms, and when there are five or more number of failure domains present in the deployment, you can configure Ceph monitor counts based on the number of racks or zones. An alert is displayed in the notification panel or Alert Center of the OpenShift Web Console to indicate the option to increase the number of Ceph monitor counts. You can use the **Configure** option in the alert to configure the Ceph monitor counts. For more information, see Resolving low Ceph monitor count alert .

Verification

- To verify the final Status of the installed storage cluster:

    a. In the OpenShift Web Console, navigate to **Storage → Data Foundation → Storage System → ocs-storagecluster**.

    b. Verify that **Status** of **StorageCluster** is **Ready** and has a green tick mark next to it.

- To verify if flexible scaling is enabled on your storage cluster, perform the following steps:

    1. In the OpenShift Web Console, navigate to **Storage → Data Foundation → Storage System → ocs-storagecluster**.

    2. From the Action menu, select **Edit Storage System**.

    3. In the YAML tab, search for the keys **flexibleScaling** in **spec** section and **failureDomain** in **status** section. If **flexible scaling** is true and **failureDomain** is set to host, flexible scaling feature is enabled.

```
spec:
flexibleScaling: true
[…]
status:
failureDomain: host
```

- To verify that all the components for OpenShift Data Foundation are successfully installed, see Verifying your OpenShift Data Foundation deployment .

Additional resources

- Scaling Storage guide

CHAPTER 3. VERIFYING OPENSHIFT DATA FOUNDATION DEPLOYMENT FOR INTERNAL MODE

Use this section to verify that OpenShift Data Foundation is deployed correctly.

1. Verify the state of the pods.
2. Verify that the OpenShift Data Foundation cluster is healthy .
3. Verify that the Multicloud Object Gateway is healthy .
4. Verify that the OpenShift Data Foundation specific storage classes exist .

3.1. VERIFYING THE STATE OF THE PODS

To determine if OpenShift Data Foundation is deployed successfully, you can verify that the pods are in **Running** state.

Procedure

1. Click **Workloads → Pods** from the left pane of the OpenShift Web Console.
2. Select **openshift-storage** from the **Project** drop-down list.

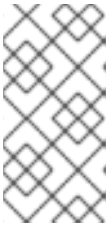


NOTE

If the **Show default projects** option is disabled, use the toggle button to list all the default projects.

For more information on the expected number of pods for each component and how it varies depending on the number of nodes, see ???.

3. Verify that the following pods are in running and completed state by clicking the **Running** and the **Completed** tabs:



NOTE

The available pods depend on the cluster configuration. When the cluster is deployed as a standalone Multicloud Object Gateway, the **rook-ceph-operator-*** pods are not available. Similarly, when the cluster is deployed without the Multicloud Object Gateway, **noobaa-*** pods are not available.

Pods corresponding to OpenShift Data Foundation cluster

Component	Corresponding pods

Component	Corresponding pods
OpenShift Data Foundation Operator	• **ocs-operator-*** (1 pod on any storage node) • **ocs-metrics-exporter-*** (1 pod on any storage node) • **odf-operator-controller-manager-*** (1 pod on any storage node) • **odf-console-*** (1 pod on any storage node) • **csi-addons-controller-manager-*** (1 pod on any storage node)
Rook-ceph Operator	**rook-ceph-operator-*** (1 pod on any storage node)
OpenShift Data Foundation Client Operator	• **ocs-client-operator-console-*** (1 pod on any storage node) • **ocs-client-operator-controller-manager-*** (1 pod on any storage node)
UX Backend	**ux-backend-server-*** (1 pod on any storage node)
Multicloud Object Gateway	• **noobaa-operator-*** (1 pod on any storage node) • **noobaa-core-*** (1 pod on any storage node) • **noobaa-db-pg-cluster-1** and noobaa-db-pg-cluster-2 (2 instances of MCG DB pod on any storage node) • **noobaa-endpoint-*** (1 pod on any storage node) • **cnpg-controller-manager-*** (1 pod on any storage node)
MON	**rook-ceph-mon-*** (3 pods on each storage node)
MGR	**rook-ceph-mgr-*** (2 pods distributed across storage nodes)

Component	Corresponding pods
MDS	**rook-ceph-mds-ocs-storagecluster-cephfilesystem-*** (2 pods distributed across storage node)
RGW	**rook-ceph-rgw-ocs-storagecluster-cephobjectstore-*** (1 pod on any storage node)
CSI	• **cephfs** ◦ **openshift-storage.cephfs.csi.ceph.com-ctrlplugin-*** (2 pods distributed across storage nodes) ◦ **openshift-storage.cephfs.csi.ceph.com-nodeplugin-*** (1 pod on each storage node) • **nfs** ◦ **openshift-storage.nfs.csi.ceph.com-ctrlplugin-*** (2 pods distributed across storage nodes) ◦ **openshift-storage.nfs.csi.ceph.com-nodeplugin-*** (1 pod on each storage node) • **rbd** ◦ **openshift-storage.rbd.csi.ceph.com-ctrlplugin-*** (2 pods distributed across storage nodes) ◦ **openshift-storage.rbd.csi.ceph.com-nodeplugin-*** (1 pod on each storage node)
rook-ceph-crashcollector	**rook-ceph-crashcollector-*** (1 pod on each storage node)
rook-ceph-exporter	**rook-ceph-exporter-worker-*** (1 pod on each storage node)

Component	Corresponding pods
OSD	• **rook-ceph-osd-*** (1 pod for each device) • **rook-ceph-osd-prepare-*** (1 pod for each device)

3.2. VERIFYING THE OPENSHIFT DATA FOUNDATION CLUSTER IS HEALTHY

Verify that the OpenShift Data Foundation cluster on IBM Power Systems is healthy after deployment by checking the storage system status and data resiliency indicators in the OpenShift Web Console.

Procedure

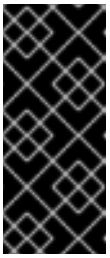
1. In the OpenShift Web Console, click **Storage → Data Foundation**.
2. Click the **Storage Systems** tab and then click on **ocs-storagecluster**.
3. In the **Status card** of Block and File dashboard under Overview tab, verify that both *Storage Cluster* and *Data Resiliency* has a green tick mark.
4. In the **Details card**, verify that the cluster information is displayed.
   For more information on the health of the OpenShift Data Foundation cluster using the Block and File dashboard, see Monitoring OpenShift Data Foundation.

3.3. VERIFYING THE MULTICLOUD OBJECT GATEWAY IS HEALTHY

Verify that the Multicloud Object Gateway is healthy after deployment to confirm that the object service and data resiliency are operational and that the NooBaa database has a recent backup.

Procedure

1. In the OpenShift Web Console, click **Storage → Data Foundation**.
2. In the **Status** card of the **Overview** tab, click **Storage System** and then click the storage system link from the pop up that appears.
   a. In the **Status card** of the **Object** tab, verify that both *Object Service* and *Data Resiliency* have a green tick.
   b. In the **Details** card, verify that the MCG information is displayed.
   For more information on the health of the OpenShift Data Foundation cluster using the object service dashboard, see Monitoring OpenShift Data Foundation.



IMPORTANT

To avoid data loss, it is recommended to take a backup of NooBaa DB PVC regularly. If NooBaa DB fails and cannot be recovered, then you can revert to the latest backed-up version. For instructions on backing up your NooBaa DB, follow the steps in the knowledgebase article, Perform a One-Time Backup of the Database for the Multicloud Object Gateway.

3.4. VERIFYING THAT THE SPECIFIC STORAGE CLASSES EXIST

Verify that the expected OpenShift Data Foundation storage classes are present after deployment.

Procedure

1. Click **Storage → Storage Classes**from the left pane of the OpenShift Web Console.
2. Verify that the following storage classes are created with the OpenShift Data Foundation cluster creation:
   - **ocs-storagecluster-ceph-rbd**
   - **ocs-storagecluster-cephfs**
   - **openshift-storage.noobaa.io**
   - **ocs-storagecluster-ceph-rgw**

CHAPTER 4. DEPLOY STANDALONE MULTICLOUD OBJECT GATEWAY

Deploying only the Multicloud Object Gateway component with the OpenShift Data Foundation provides the flexibility in deployment and helps to reduce the resource consumption. After deploying the MCG component, you can create and manage buckets using MCG object browser. For more information, see Creating and managing buckets using MCG object browser .

Use this section to deploy only the standalone Multicloud Object Gateway component, which involves the following steps:

- Installing the Local Storage Operator.
- Installing Red Hat OpenShift Data Foundation Operator
- Creating standalone Multicloud Object Gateway



IMPORTANT

To avoid data loss, it is recommended to take a backup of NooBaa DB PVC regularly. If NooBaa DB fails and cannot be recovered, then you can revert to the latest backed-up version. For instructions on backing up NooBaa DB, follow the steps in the knowledgebase article, Backup and Restore for Multicloud Object Gateway database (NooBaa DB).

4.1. INSTALLING LOCAL STORAGE OPERATOR

Use this procedure to install the Local Storage Operator from the Software Catalog before creating OpenShift Data Foundation clusters on local storage devices.

Procedure

1. Log in to the OpenShift Web Console.
2. Click **Ecosystem → Software Catalog**
3. Type **local storage** in the **Filter by keyword...** box to find the **Local Storage Operator** from the list of operators and click on it.
4. Set the following options on the **Install Operator** page:
    a. Update channel as **stable**.
    b. Installation Mode as **A specific namespace on the cluster**
    c. Installed Namespace as **Operator recommended namespace openshift-local-storage**.
    d. Approval Strategy as **Automatic**.
5. Click **Install**.

Verification

- Verify that the Local Storage Operator shows a green tick indicating successful installation.

4.2. INSTALLING RED HAT OPENSHIFT DATA FOUNDATION OPERATOR

You can install Red Hat OpenShift Data Foundation Operator using the Red Hat OpenShift Container Platform Software Catalog.

For information about the hardware and software requirements, see Planning your deployment.

Prerequisites

- Access to an OpenShift Container Platform cluster using an account with **cluster-admin** and Operator installation permissions.
- You must have at least three worker nodes in the Red Hat OpenShift Container Platform cluster.



IMPORTANT

- When you need to override the cluster-wide default node selector for OpenShift Data Foundation, you can use the following command in the command line interface to specify a blank node selector for the **openshift-storage** namespace (create openshift-storage namespace in this case):

```
$ oc annotate namespace openshift-storage openshift.io/node-selector=
```

- Taint a node as **infra** to ensure only Red Hat OpenShift Data Foundation resources are scheduled on that node. This helps you save on subscription costs. For more information, see How to use dedicated worker nodes for Red Hat OpenShift Data Foundation chapter in the Managing and Allocating Storage Resources guide.

Procedure

1. Log in to the OpenShift Web Console.
2. Click **Ecosystem → Software Catalog**
3. Scroll or type **OpenShift Data Foundation** into the **Filter by keyword** box to find the **OpenShift Data Foundation** Operator.
4. Click **Install**.
5. Set the following options on the **Install Operator** page:
   a. Update Channel as **stable-4.22**.
   b. Installation Mode as **A specific namespace on the cluster**
   c. Installed Namespace as **Operator recommended namespace openshift-storage**. If Namespace **openshift-storage** does not exist, it is created during the operator installation.
6. Select **Approval Strategy** as **Automatic** or **Manual**.
   If you select **Automatic** updates, then the Operator Lifecycle Manager (OLM) automatically upgrades the running instance of your Operator without any intervention.

If you select **Manual** updates, then the OLM creates an update request. As a cluster administrator, you must then manually approve that update request to update the Operator to a newer version.

7. Ensure that the **Enable** option is selected for the **Console plugin**.
8. Click **Install**.

Verification

- Verify that the **OpenShift Data Foundation** Operator shows a green tick indicating successful installation.
- After the operator is successfully installed, the web console automatically reloads to apply the changes. During this process, a temporary error message might appear on the page and this is expected and disappears after the refresh completes.
    - In the Web Console, navigate to **Storage** and verify if **Data Foundation** is available.

4.3. CREATING STANDALONE MULTICLOUD OBJECT GATEWAY ON IBM POWER

You can create only the standalone Multicloud Object Gateway component while deploying OpenShift Data Foundation.

Prerequisites

- Ensure that the OpenShift Data Foundation Operator is installed.
- The Multicloud Object Gateway database uses ReadWriteOncePod (RWOP) access mode to safeguard against node failures. If the underlying CSI driver does not support RWOP, the access mode must be manually changed to ReadWriteOnce (RWO) before deployment. For more information, see the knowledgebase article How to change NooBaa DB access mode from RWOP to RWO in standalone MCG deployments?.
- (For deploying using local storage devices only) Ensure that Local Storage Operator is installed.

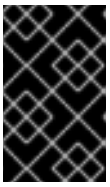
To identify storage devices on each node, refer to Finding available storage devices.

Procedure

1. Log into the OpenShift Web Console.
2. In **openshift-local-storage** namespace, click **Ecosystem → Installed Operators** to view the installed operators.
3. Click the **Local Storage** installed operator.
4. On the **Operator Details** page, click the **Local Volume** link.
5. Click **Create Local Volume**.
6. Click on **YAML view** for configuring Local Volume.
7. Define a **LocalVolume** custom resource for filesystem PVs using the following YAML.

```
apiVersion: local.storage.openshift.io/v1
kind: LocalVolume
metadata:
  name: localblock
  namespace: openshift-local-storage
spec:
  logLevel: Normal
  managementState: Managed
  nodeSelector:
    nodeSelectorTerms:
      - matchExpressions:
          - key: kubernetes.io/hostname
            operator: In
            values:
              - worker-0
              - worker-1
              - worker-2
  storageClassDevices:
    - devicePaths:
        - /dev/sda
      storageClassName: localblock
      volumeMode: Filesystem
```

The above definition selects **sda** local device from the **worker-0**, **worker-1** and **worker-2** nodes. The **localblock** storage class is created and persistent volumes are provisioned from **sda**.



IMPORTANT

Specify appropriate values of nodeSelector as per your environment. The device name should be same on all the worker nodes. You can also specify more than one devicePaths.

8. Click **Create**.
9. In the OpenShift Web Console, navigate to **Storage → StorageCluster**.
10. On the Welcome page, click **Configure Data Foundation**.
11. Select **Setup Multicloud Object Gateway**.
12. Select the **Use an existing StorageClass** option for **Backing storage type**.
    a. Select the **Storage Class** that you used while installing LocalVolume.
13. Click **Next**.
14. Optional: In the **Security** page, select the **Connect to an external key management service** checkbox. This is optional for cluster-wide encryption.
    a. From the **Key Management Service Provider** drop down list, either select **Vault** or **Thales CipherTrust Manager (using KMIP)**. If you selected **Vault**, go to the next step. If you selected **Thales CipherTrust Manager (using KMIP)**, go to step iii.
    b. Select an **Authentication Method**.

Using Token authentication method

- Enter a unique **Connection Name**, host **Address** of the Vault server ('https://<hostname or ip>'), **Port** number and **Token**.
- Expand **Advanced Settings** to enter additional settings and certificate details based on your **Vault** configuration:
  - Enter the Key Value secret path in **Backend Path** that is dedicated and unique to OpenShift Data Foundation.
  - Optional: Enter **TLS Server Name** and **Vault Enterprise Namespace**.
  - Upload the respective PEM encoded certificate file to provide the **CA Certificate**, **Client Certificate** and **Client Private Key** .
  - Click **Save** and skip to step iv.

Using Kubernetes authentication method

- Enter a unique Vault **Connection Name**, host **Address** of the Vault server ('https://<hostname or ip>'), **Port** number and **Role** name.
- Expand **Advanced Settings** to enter additional settings and certificate details based on your **Vault** configuration:
  - Enter the Key Value secret path in the **Backend Path** that is dedicated and unique to OpenShift Data Foundation.
  - Optional: Enter **TLS Server Name** and **Authentication Path** if applicable.
  - Upload the respective PEM encoded certificate file to provide the **CA Certificate**, **Client Certificate**, and **Client Private Key**.
  - Click **Save** and skip to step iv.

c. To use **Thales CipherTrust Manager (using KMIP)** as the KMS provider, follow the steps below:

   i. Enter a unique **Connection Name** for the Key Management service within the project.

   ii. In the **Address** and **Port** sections, enter the IP of Thales CipherTrust Manager and the port where the KMIP interface is enabled. For example:

   - **Address**: 123.34.3.2
   - **Port**: 5696

   iii. Upload the **Client Certificate**, **CA certificate**, and **Client Private Key**.

   iv. If StorageClass encryption is enabled, enter the Unique Identifier to be used for encryption and decryption generated above.

   v. The **TLS Server** field is optional and used when there is no DNS entry for the KMIP endpoint. For example, **kmip_all_<port>.ciphertrustmanager.local**.

d. Select a **Network**.

e. Click **Next**.

15. In the **Review and create** page, review the configuration details:
To modify any configuration settings, click **Back**.

16. Click **Create StorageSystem**.

Verification

Verifying that the OpenShift Data Foundation cluster is healthy

1. In the OpenShift Web Console, click **Storage → Data Foundation**.

2. Click the **Storage Systems** tab and then click on **ocs-storagecluster**.

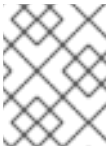
   a. In the **Status card** of the **Object** tab, verify that both *Object Service* and *Data Resiliency* have a green tick.

   b. In the **Details** card, verify that the MCG information is displayed.

Verifying the state of the pods

1. Click **Workloads → Pods** from the OpenShift Web Console.

2. Select **openshift-storage** from the **Project** drop-down list and verify that the following pods are in **Running** state.



NOTE

If the **Show default projects** option is disabled, use the toggle button to list all the default projects.

Component	Corresponding pods
OpenShift Data Foundation Operator	• **ocs-operator-*** (1 pod on any storage node) • **ocs-metrics-exporter-*** (1 pod on any storage node) • **odf-operator-controller-manager-*** (1 pod on any storage node) • **odf-console-*** (1 pod on any storage node) • **csi-addons-controller-manager-*** (1 pod on any storage node)
Rook-ceph Operator	**rook-ceph-operator-*** (1 pod on any storage node)

Component	Corresponding pods
Multicloud Object Gateway	• **noobaa-operator-*** (1 pod on any storage node) • **noobaa-core-*** (1 pod on any storage node) • **noobaa-db-pg-cluster-1** and noobaa-db-pg-cluster-2 (2 instances of MCG DB pod on any storage node) • **noobaa-endpoint-*** (1 pod on any storage node) • **cnpg-controller-manager-*** (1 pod on any storage node) • **noobaa-default-backing-store-noobaa-pod-*** (1 pod on any storage node)

CHAPTER 5. VIEW OPENSHIFT DATA FOUNDATION TOPOLOGY

The topology provides a visual representation of the OpenShift Data Foundation storage cluster at multiple levels of abstraction and enables interaction with each layer. It illustrates how the cluster components are organized and related, while offering visibility into Object Storage Devices (OSDs), including their count, type, and status.

Procedure

1. On the OpenShift Web Console, navigate to **Storage → Data Foundation → Topology**.
   The view shows the storage cluster and the topology groupings inside it. You can see the nodes depicted by circular entities within the topology groupings, which are indicated by dotted lines. The topology groupings are based on the cluster's topology configuration (for example, racks in bare metal or vSphere clusters, or zones in cloud-based clusters). The label of each item or resource contains basic information such as status and health or indication for alerts.

2. Choose a node to view node details on the right-hand panel. You can also access resources or deployments within a node by clicking on the search/preview decorator icon.
   The node details panel displays information about the OSDs running on that node, including:

   - Number of OSDs on the node
   - OSD disk type (for example, SSD, HDD)
   - OSD status (for example, Running, Error, CrashLoopBackOff)

3. To view deployment details

   a. Click the preview decorator on a node. A modal window appears above the node that displays all of the deployments associated with that node along with their statuses.

   b. Click the **Back to main view** button in the model's upper left corner to close and return to the previous view.

   c. Select a specific deployment to see more information about it. All relevant data is shown in the side panel.

4. Click the **Resources** tab to view the pods information. This tab provides a deeper understanding of the problems and offers granularity that aids in better troubleshooting.

5. Click the **OSD information** tab to view detailed information about the OSDs. This tab provides more comprehensive OSD-specific details, including:

   - OSD pod name and status
   - Disk device associated with each OSD
   - Device class (for example, SSD, HDD)
   - Resource utilization metrics

6. Click the pod links to view the pod information page on OpenShift Container Platform. The link opens in a new window.

CHAPTER 6. UNINSTALLING OPENSHIFT DATA FOUNDATION

6.1. UNINSTALLING OPENSHIFT DATA FOUNDATION IN INTERNAL MODE

Uninstalling OpenShift Data Foundation in internal mode permanently removes the storage cluster, all associated PVCs, and the operator.

Procedure

1. To uninstall OpenShift Data Foundation in internal mode, refer to the knowledgebase article on Uninstalling OpenShift Data Foundation.